

**Sessional 1 (Nov (2022))
B.Tech (ME) - 5th Sem
DME-1 (PCC-ME-503/21)**

Duration: 90 Minutes

Max. Marks: 15

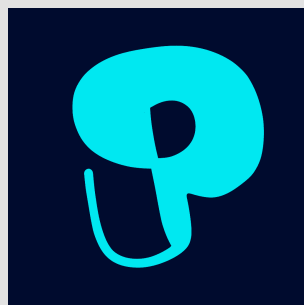
Instructions:

- Attempt all questions
- All questions carry equal marks

Q1 What are the factors influencing the selection of F.O.S. ? (5) [CO1]

Q2 Describe the concept of fatigue failures in detail. (5) [CO1]

Q3 Two flat plates subjected to a tensile force P are connected together by means of double-strap butt joint. The force P is 250 kN and the width of the plate w is 200 mm. The rivets and plates are made of the same steel and the permissible stresses in tension, compression and shear are 70, 100 and 60 N/mm² respectively. Calculate:
(i) the diameter of the rivets;
(ii) the thickness of the plates;
(iii) the dimensions of the seam, viz., p , p_t and m ;
(iv) the efficiency of the joint (5) [CO2]



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